STUDIES IN SPATIAL LEARNING. VI. PLACE ORIENTATION AND DIRECTION ORIENTATION

BY BENBOW F. RITCHIE[1]

Swarthmore College

INTRODUCTION

In the first and second articles in this series (3, 4), it was pointed out that the purpose of this series was to develop some of the important implications of Tolman's 'expectancy theory.' It was also pointed out that when this theory is applied to problems of maze learning it is very similar to Woodworth's 'place-learning hypothesis' (6, 7). Both of these theories or hypotheses emphasize the idea that what is learned in any spatial problem is not simply a specific response sequence, but is *something* about the spatial relations between the various objects in the rat's environmental field. Neither of these hypotheses, however, specify *what* is learned about these spatial relations, and because of this lack of specificity these hypotheses are not sufficiently precise to permit experimental investigation.

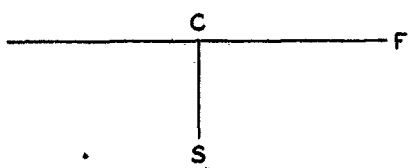


FIG. 1. Schematic drawing of simple T-maze

Consider now a rat which has been trained on a simple T-maze (see Fig. 1) so that whenever the rat is hungry and is put at S he runs quickly up to C and turns without hesitation to the right towards F. Here is an extremely simple example of spatial learning. Now how do Tolman's and Woodworth's theories of such learning differ from what has been called 'the orthodox S-R theory of maze-learning'? Primarily the difference between their theories and the S-R theory seems to be that the latter kind of theory regards such learning as the acquisition of a single disposition, namely, the disposition to take the

[1] This investigation is part of a co-ordinated research program of the Psychological Laboratory at the University of California. The writer is deeply indebted to Professor E. C. Tolman for generous advice and aid. The present paper is a portion of a dissertation presented to the Faculty of the Graduate School of the University of California in partial fulfillment of the requirement for the degree of Doctor of Philosophy.

practiced path, SCF. Tolman and Woodworth, on the other hand, suggest that not only is this disposition acquired as a result of the training, but other dispositions as well. Now, whether other dispositions are acquired during training on a particular path is, of course, an empirical question, and it is the primary purpose of this series of studies to answer this question, and to discover something about the nature of such *auxiliary dispositions* if they do in fact exist.[2]

In the present study we will confine our discussion to two kinds of auxiliary dispositions, (1) an auxiliary place disposition, and (2) an auxiliary direction disposition. We will first consider the meaning of these terms, and then discuss the evidence that we have for believing that such auxiliary dispositions are sometimes acquired.

1. *Auxiliary place dispositions.*—If after training on a simple T-maze, (1) the path from S to C is blocked (see Fig. 2) and (2) several

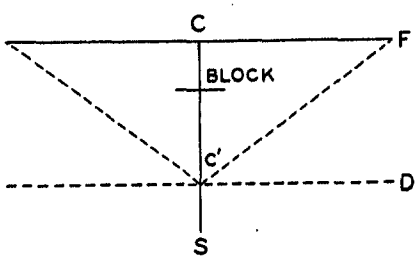


FIG. 2. Schematic drawing of test situation for auxiliary dispositions. S = original starting place. C = original choice point. F = original food place. C′ = new choice point.

new paths are offered to the rat, one of which (i.e., C′F) points directly to the original food place, *then*, if under these changed conditions the rat chooses path C′F, we shall say that the rat has acquired an auxiliary place disposition.

2. *Auxiliary direction dispositions.*—If under exactly the same conditions, the rat then chooses a path (i.e., path C′D) which runs parallel to the original true path (i.e. path CF) and points in the same direction, we shall say that the rat has acquired an auxiliary direction disposition.

The experiment reported in the first article in this series (3) presented evidence that auxiliary place dispositions are sometimes acquired by rats.

[2] I shall call the disposition to take the practiced path 'the primary disposition,' and all others 'auxiliary dispositions.' It should be emphasized however that these terms are merely introduced for the sake of abbreviation, and not because I believe that the disposition to take the practiced path is more fundamental than the other dispositions. In fact it may be discovered that the auxiliary dispositions are more basic than the primary disposition in the sense that the acquisition of the auxiliary ones is a necessary condition for the acquisition of the primary disposition.

Briefly the procedure was this. A group of 56 maze-wise rats was trained in a new spatial problem which involved running over a table-top, through an alley, and then down an elevated path which ran at right angles to the starting path. When these rats had received five such training trials, the original path was blocked at the end of the alley, and 18 elevated paths were attached to the circular table-top radiating from its edge at intervals of 10 degrees. On the test trial *all* of the rats ran down the alley to the block, and then returned to the table top. They all explored the table-top and the initial segments of the attached paths. After a great amount of such exploration all but three of them chose one of these new paths and ran out to the end of the path. Thirty-six percent of the rats in this group chose the path which pointed most directly towards the *former* food location. The rest of the rats in the group were distributed randomly on the remaining 17 paths. We concluded from these results that 36 percent of our rats had acquired an auxiliary place disposition, and we called this disposition 'an expectation of food at location L.'

The evidence for the existence of auxiliary direction dispositions is not as simple or direct as the evidence for auxiliary place dispositions. In fact, it was only after observing the behavior of some rats in the first experiment reported in the third article in this series (5) that we hit upon the idea of auxiliary direction dispositions. In this experiment we observed that the rats in the control group after having been trained on a simple T-maze, continued to run in the same direction (i.e., towards the west wall) no matter where the maze was put in the room, as long as the path from the choice point to the food (i.e., C'F) was parallel to the original CF path. As a consequence we concluded that this behavior *might* be the result of having acquired an auxiliary direction disposition.

This, then, is all the evidence we have at present for the existence of such direction dispositions. Although this evidence is far from overwhelming, it was sufficiently cogent to suggest that further research on this point might be fruitful.

We turn now to a new problem. It should be clear that if such auxiliary place or direction dispositions are sometimes acquired by rats, in such cases the behavior of the rats is a function not only of the intra-maze environment, but of the extra-maze environment as well. Furthermore the extra-maze cues in such cases seem to play a peculiar role, a role which is different from the role usually attributed to the more proximal intra-maze cues. Thus, consider a rat on the test trial in which he selects a completely new path (one which points towards the original food place, or one which points in the same direction as the original true path). In such a case the extra-maze cues are 'received proximally' from a wholly new perspective, and yet somehow they seem to be recognized and interpreted in a way that is appropriate to the objective spatial relation of the objects in the environmental field. By this I mean that the rats are able to choose new paths which point towards the same place in the objective environment, or point in

the same direction, despite the fact that the perspectival relations of the rats to the extra-maze cues are entirely novel.

The question now arises as to whether this 'grasping of objective spatial relations' (despite changes in the immediate proximal perspective) is restricted to the original, practiced starting place or starting path, or whether when acquired in one starting place the auxiliary direction or place dispositions are of a sort to generalize to new starting places in the objective environmental field. The following experiment was designed to answer this question. First, a group of rats was trained on a simple T-maze from a single starting place, and then after this maze had been learned, we determined what auxiliary place or direction dispositions the rats had acquired, if any, from a new starting place. For reasons which will be explained in the conclusion of this article, the *new* starting place which was used was a place on the opposite side of the choice point of the T-maze, on the same axis as the original starting path, and at the same distance from the choice point as the original starting place.

Animals

Fifty pigmented male rats approximately three months old were used in this experiment. These rats were MxM's drawn from the Tryon stock in the rat colony of the University of California. None of them had had any previous maze training. They were put on a 24-hour wet-food maintenance schedule five days before the beginning of the preliminary training. During the training they were run between 8 and 11 P.M.

Apparatus

Figs. 3 and 4 present diagrams of the apparatus which were used. In Fig. 3 we see the apparatus used in the preliminary training. It consisted of an unpainted wooden circular table

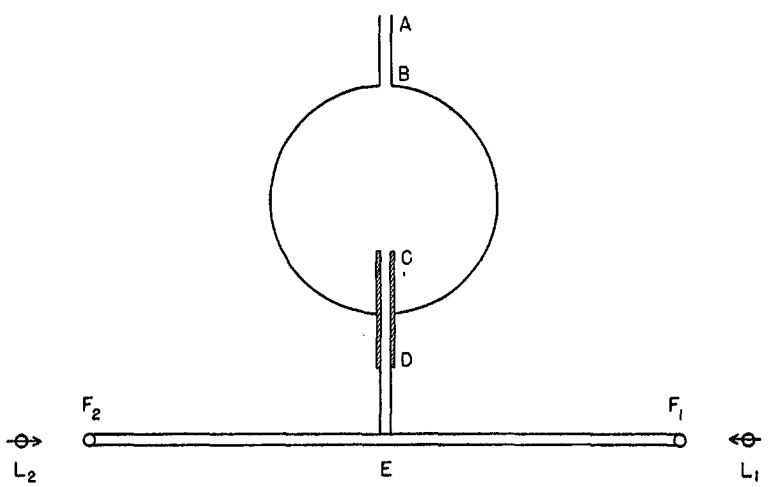


Fig. 3. Apparatus used in preliminary training

top 91 cm. in diameter and several elevated wooden paths painted black. The path AB was 27 cm. in length and was used as a starting path. The path CE was 75 cm. in length, while path F_1F_2 was 2.5 meters long. At each end of the F_1F_2 path were small paper food cups three cm. in diameter and three cm. in depth. Twenty cm. beyond each end of the F_1F_2 path were small lights L_1 and L_2, held in a socket so that the beam of light was directed on the food cup below. These lights contained 15-watt bulbs which were covered with the paper carton in which they were sold. They were 20 cm. above the level of the F_1F_2 path. The only other illumination in the room was located at L_3 (see Fig. 5A). At this place was an ordinary desk lamp containing a 75 watt bulb. The reflector of this lamp was turned in such a way that its light was directed on the white wall of the room at an angle of 45 degrees. Thus, this corner of the room (see Fig. 5A) was brightly lighted while the rest of the room was in deep shadow. Fastened to the sides of the path DE were two pieces of unpainted plywood, which were 47 cm. in height and 47 cm. in length. These formed an alley which began 20 cm. from the center of the table-top and ended 27 cm. away from the choice point E.

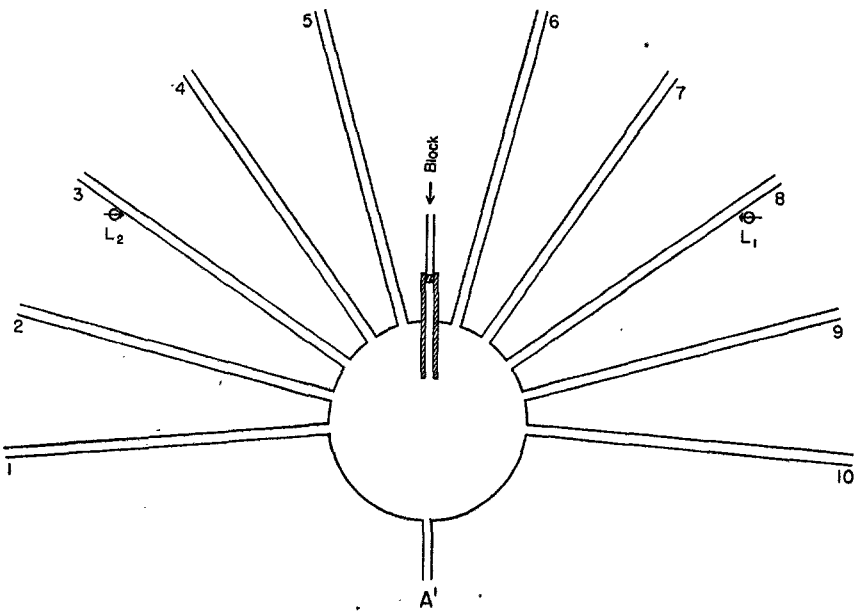


FIG. 4. Apparatus used in test trial

In Fig. 4 we see the apparatus used in the test trial. This consisted of the same starting path, circular table-top, alley on path CE, and lights at L_1, L_2, and L_3. But the path F_1F_2 with food cups was removed, and a block was placed at the end of the alley on path CD. Then 10 paths 1.5 meters in length were placed around the table-top. These paths began at a point 95 degrees to the right of path CE and radiated in a counterclockwise fashion, each path being placed 20 degrees to the left of its neighbor.[8]

As can be seen in Fig. 5A the choice point E of the maze used in the preliminary training was in the center of the room and the F_1F_2 path ran parallel to the north and south walls. The apparatus used in the test trial was set exactly opposite to the other apparatus (see Fig 5B) so that the starting point A′ was now the same distance from the south wall as it had been from the north wall in the preliminary training. The positions of the lights remained however unchanged.

The rats lived in cages in another room and were carried in their cages from their living room to the experimental room for their training. Their cages were then put in a rack on the same axis

[8] There is one exception to this. Path 5 was 30 degrees to the left of path 6.

as that of the starting path and the choice point of the maze used in the preliminary training (see Fig. 5A). On the test trial they were put in the same rack. The *E* carried the rats by hand, put them on the apparatus by hand, and then sat on a stool beside the rack where the rats were kept.

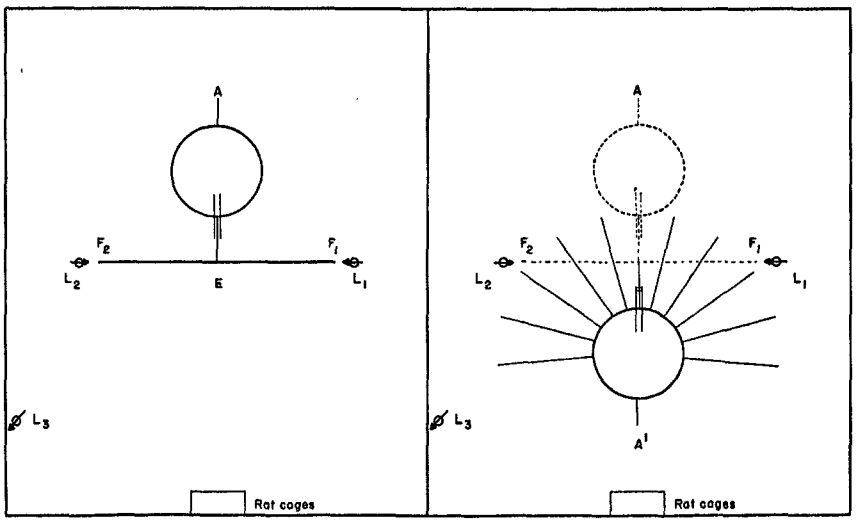


FIG. 5. Floor plans showing positions of apparatus. Positions during preliminary training in 5A. Positions during test trial in 5B.

METHOD

Pre-test procedures.—On Days 1 and 2 the rats were trained in another room to accustom them to turning on straight paths and to eating from paper food cups.

On Day 3 a block was placed at D on the apparatus in Fig. 3 and the rats were given six trial of running from F_1F_2.[4] On each of these trials the rats were allowed to eat at F_1 for 20 sec. and then were returned to the cage in the rack.

On Day 4 the rats were given two trials from D to F_1 (with a block at D) and two more trials from D to F_1 (without the block at D). Then they were give two trials through the alley from C to F_1.

On Day 5 they were given two more trials through the alley from C to F_1. They were then given four trials from the start A to F_1.

On Days 6 and 7 they were given eight daily trials from A to F_1. Thus by the end of training on Day 7, each rat had run 20 times from A to F_1.

Test procedures.—On Day 8 one test trial was given. The apparatus was changed to that represented in Fig. 4, and placed in the room as pictured in Fig. 5B. Each rat was started at A' and allowed to explore the blocked alley, the table-top, and the various alternative paths which radiated from it. If any rat put both front feet on any path but did not run out more than two feet on the path[5] this was recorded as a 'partial choice.' A rat's trial ended as soon as it had chosen one of the paths and had run out on it at least four feet. If any rat took more than seven min. to make such a choice, then it was removed.

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[4] Twenty-three of the rats were trained to find food at F_1. This group will be the 'F_1 group.' Twenty-seven of the rats were trained to find food at F_2, and they were called the 'F_2 group.' In the following description of the training procedure we will consider only the F_1 group. The analogous kind of training was of course given to the F_2 group.

[5] It was discovered in the first short-cut experiment (3) that any rat which ran more than two feet out on a path ran all the way out. The same behavior was observed in this experiment.

Results

On the test trial, eight of the 50 rats were discarded because they failed to make a choice after seven min. After having explored the paths and table-top, all of these eight rats returned to the center of the table and refused to move from there except to return either to the alley or to the starting path.

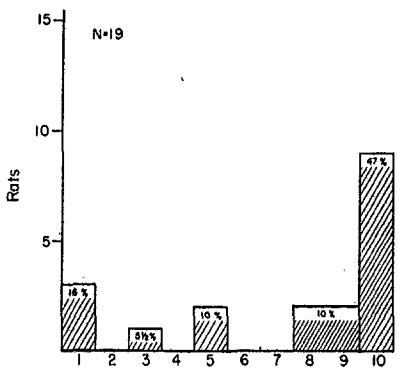


Fig. 6. Percentage of path choice in the F_1 group

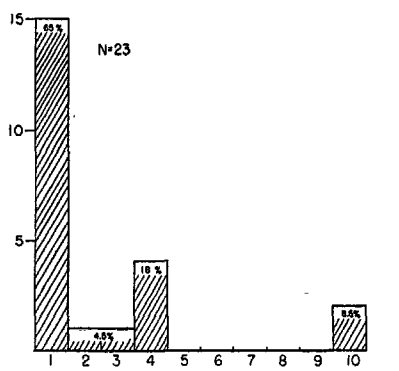


Fig. 7. Percentage of path choice in the F_2 group

Fig. 6 shows the distribution of choices made by the remaining rats in the F_1 group. Inspection of this distribution shows that only 10.5 percent of the rats in the F_1 group chose path 8, the one which points most directly towards F_1. The distribution of the choices in the F_2 group (see Fig. 7) is similar; only 4.5 percent of the rats in this group chose path number 3, the one which points towards F_2.

While only a few of the rats chose the most direct path to the original food place, the path chosen most frequently by both groups was the last path on the side of the table which faced the original food place. Thus, 47 percent of the F_1 group chose path number 10, and 65 percent of the F_2 group chose path number 1.

Now, since the problem set for the F_1 group was the mirror image of that set for the F_2 group, the choices of both groups can be combined as in Fig. 8, where A is the choice of the last path on the side facing the former food place (e.g. in the F_1 group path 10, and in the F_2 group path 1), C is the most direct path to the former food place, and so forth. When the results of both groups are combined in this way we see that 57 percent of the 42 rats chose the last path on the food side, while the distribution on the other paths never exceeded 11 percent.

In Fig. 9 we see the distribution of partial choices (i.e. in which the rats did not run out more than two feet) of the F_1 group, and in Fig. 10

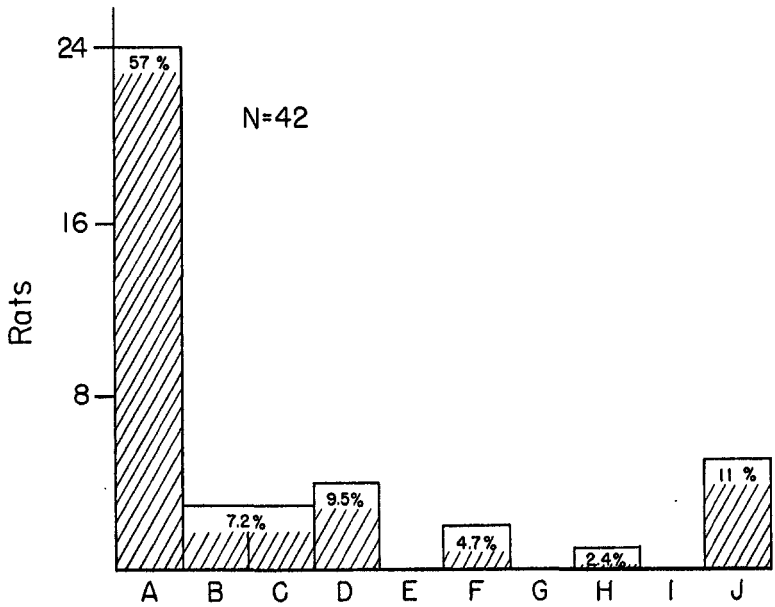


FIG. 8. Percentage of path choice in both groups

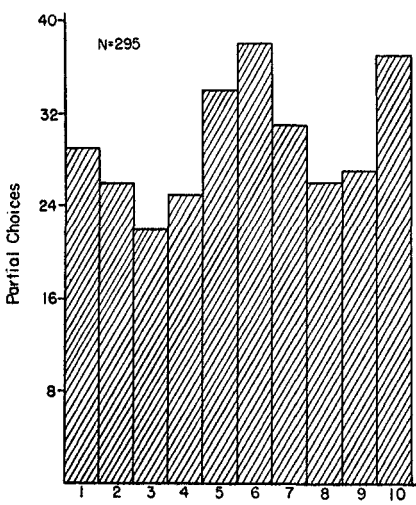


FIG. 9. Partial choices in F_1 group

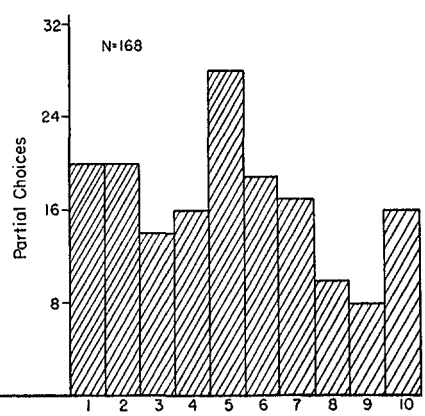


FIG. 10. Partial choices in F_2 group

we see the distribution of partial choices for the F_2 group. These two distributions show that the rats in both groups explored all paths about equally before making a choice. There was, however, one difference between the two groups. The F_1 group made approximately twice as many partial choices as the F_2 group. The mean number of partial choices for the F_1 group was 15.5 with a standard deviation of

8.5, while the mean number of partial choices for the F_2 group was 7.2 with a standard deviation of 5.5. The critical ratio of the difference between these two means was 11.2.

Conclusion

The results of this experiment show that auxiliary direction dispositions sometimes generalize to new starting places in the objective environmental field. Further, it is clear that in *our* experimental situation auxiliary place dispositions do not. We cannot, however, conclude from these results that auxiliary place dispositions never generalize to new starting places, since we know nothing, as yet, about the environmental factors which favor place as compared with those factors which favor direction dispositions.

Far from being unexpected, results such as these were predicted from our interpretation of the results of other experiments in this series. In fact, the notion that some kind of auxiliary disposition was generalized to other starting places had been suggested by the results of the early Honzik and Tolman (2) experiment, 'The perception of spatial relations by the rat.' In this experiment rats were trained on an elevated T-maze (Fig. 11) to prefer the shorter path from S_1 to F_1 and to avoid the longer path S_1 to F_2, although there was food at both F_1 and F_2.

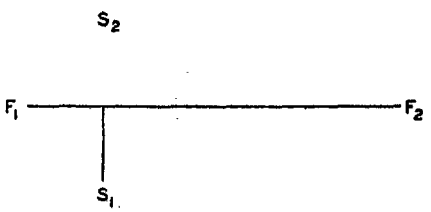


Fig. 11. Schematic drawing of maze used by Honzik and Tolman (2) in which rats were first trained to prefer the shorter path S_1 to F_1. Then, after having been shocked at F_1, they were tested from S_1 and from S_2.

After this preference was established, the rats were carried by hand and shocked in the food box at F_1. When they had recovered from this shock they were again started at S_1 and 87 percent of the animals which had been going consistently to F_1, now turned and went to F_2. Next, in order to demonstrate that the shock had not merely served to reverse their position or turn habit, another group of animals were given exactly the same training, and then after being shocked at F_1, they were tested from a *new* starting place S_2 (see Fig. 11). Again 80 percent of the rats went to F_2. Such results strongly suggest that some kind of auxiliary disposition which is acquired when the rat is practiced from S_1, generalizes or transfers to the new starting place S_2.

A simpler example of apparently the same phenomenon is provided by the experiment in the second article in this series (4). Here two groups of

rats were trained on the simple elevated maze shown in Fig. 12. Both groups of rats were hungry and were given six trials a day, three starting from S_1 and three starting from S_2. One group (the Place Group) were required to learn to go always to the same place, for example to F_1, while the other group (the Response Group) were required to learn to turn always

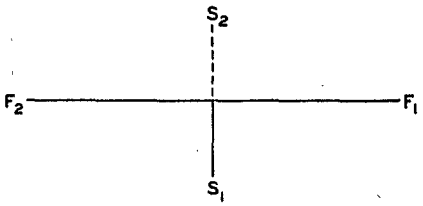


FIG. 12. Schematic drawing of maze used by Tolman, Ritchie, and Kalish (4)

to the right at the choice point. Thus, when started at S_1 members of the Response Group were required to go to F_1 and when started at S_2 to go to F_2. If auxiliary place or direction dispositions (1) transfer to new starting places, and (2) support or reinforce the related primary dispositions or dispositions to take the practiced path, then the problem for the Place Group should be easier than the problem for the Response Group.[6] The results of this experiment showed that place-learning in a situation where there were marked extra-maze cues was much easier than response-learning.

But such results do not provide unambiguous evidence for the transference of auxiliary dispositions, since other hypotheses can account for the results equally well. For example, it might be argued that in such a situation it is difficult to learn to make the same turn response repeatedly, since a situation like this makes for response variability. Thus, the Place Group which is permitted variability learns more repidly than the Response Group which is not.

In order to eliminate the possibility of such an explanation, the experiments in the third article in this series (5) were performed. In the second experiment in this third article, the Experimental Group was given the same kind of training as the Place Group in the experiment just described. The Control Group, on the other hand, was given two separate place problems. They were trained to run from S_a in one room (Room A) to F_a, and to run from S_b, in another room (Room B), where the extra-maze cues were totally different, to F_b. When started in Room A they were required to turn right, and in Room B they were required to turn left. Thus the pattern of turn responses in the experimental and control groups was identical, and the only difference between the problems set for the two groups was that the experimental group were required to go to the same food place from two different starting places, while the control group had to go to two

[6] This conclusion is of course based on the further assumption that in situations where there are marked extra-maze cues, turn dispositions (e.g. to the right) do not generalize to new starting positions. As yet we have no clear evidence that such turn dispositions are ever acquired in such situations. Blogett and McCutchan (1) suggest that when extra-maze cues are eliminated such turn dispositions may be acquired.

different food places from two different starting places. If our interpretation of the previous experiments has been correct then the one-place or experimental group should learn faster than the two-place control group. This hypothesis was confirmed by the results of this experiment. Finally this article concluded with a theoretical explanation of these results in terms of the transference of a short-cut disposition (i.e. the auxiliary place disposition).

The experiment in the present article was performed to test this theoretical explanation, and the results indicate that although there is no evidence that auxiliary place dispositions transfer to new starting places, there is good evidence that auxiliary direction dispositions do.

Summary

1. Orthodox S-R theory has seemed to argue that the only habit that is learned in a maze problem is a disposition to take the practiced path. Tolman's and Woodworth's theories, on the other hand, emphasize that other, auxiliary dispositions are also acquired.

2. Two kinds of auxiliary dispositions are herein distinguished, (1) place dispositions and (2) direction dispositions, and the evidence for the existence of such dispositions is reviewed.

3. To determine whether such auxiliary dispositions generalize to new starting places or paths, a group of 50 rats was trained on a simple T-maze similar to the one used in the short-cut experiment in the first paper in this series (3). After this training was completed, each of the rats were started from a place which was on the opposite side of the choice point of the original maze, and on the same axis as the original starting path.

4. Fifty-seven percent of the rats exhibited an auxiliary direction disposition. That is, they chose the last path on the side facing the former food place. There was no evidence that these rats also acquired an auxiliary place disposition. Only 7.2 percent of the animals chose the path that pointed directly towards the former food place.

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